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A source-free unsupervised domain adaptation method for cross-regional and cross-time crop mapping from satellite image time series

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ABSTRACT

Precise and timely information about crop types plays a crucial role in various agriculture-related applications. However, crop type mapping methods often face significant challenges in cross-regional and cross-time scenarios with high discrepancies between temporal-spectral characteristics of crops from different regions and years. Unsupervised domain adaptation (UDA) methods have been employed to mitigate the problem of domain shift between the source and target domains. Since these methods require source domain data during the adaptation phase, they demand significant computational resources and data storage, especially when large labeled crop mapping source datasets are available. This leads to increased energy consumption and financial costs. To address this limitation, we developed a source-free UDA method for cross-regional and cross-time crop mapping, capable of adapting the source-pretrained models to the target datasets without requiring the source datasets. The method mitigates the domain shift problem by leveraging mutual information loss. The diversity and discriminability terms in the loss function are balanced through a novel unsupervised weighting strategy based on mean confidence scores of the predicted categories. Our experiments on mapping corn, soybean, and the class Other from Landsat image time series in the U.S. demonstrated that the adapted models using different backbone networks outperformed their non-adapted counterparts. With CNN, Transformer, and LSTM backbone networks, our adaptation method increased the macro F1 scores by 12.9%, 7.1%, and 5.8% on average in cross-time tests and by 20.1%, 12.5%, and 8.8% on average in cross-regional tests, respectively. Additionally, in an experiment covering a large study area of 450 km × 300 km, the adapted model with the CNN backbone network obtained a macro F1 score of 92.6%, outperforming its non-adapted counterpart with a macro F1 score of 89.2%. Our experiments on mapping the same classes using Sentinel-2 image times series in France demonstrated the effectiveness of our method across different countries and sensors. We also tested our method in more diverse agricultural areas in Denmark and France containing six classes. The results showed that the adapted models outperformed the non-adapted models. Moreover, in within-season experiments, the adapted models performed better than the non-adapted models in the vast majority of weeks. These results and their comparison to those obtained by the other investigated UDA methods demonstrated the efficiency of our proposed method for both end-of-season and within-season crop mapping tasks. Additionally, our study showed that the method is modular and flexible in employing various backbone networks. The code and data are available at https://github.com/Sina-Mohammadi/SFUDA-CropMapping.

1. Introduction

Accurate information on cultivated crops plays a crucial role in food security research (Sagan et al., 2021), monitoring crop growth (Ma et al., 2022), and understanding how climate change might impact the production and nutritional quality of crops (Belgiu et al., 2023). This information is usually collected through costly and time-consuming field campaigns (Wang et al., 2020). Consequently, in recent years, there has been an increasing interest in developing efficient methods to classify crops, especially using satellite image time series that are capable of capturing the dynamics of crops over their entire growing season and across large areas.

Crop mapping has been accomplished through the implementation of both machine learning (Duro et al., 2012; Yan et al., 2021) and deep learning methods (Turkoglu et al., 2021; Pelletier et al., 2019; Mohammadi et al., 2023). Deep learning methods, in particular, achieved promising results due to their ability to learn complex patterns from remote sensing image time series (Zhong et al., 2019; Xu et al., 2020). To obtain good classification results, these methods require a large

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amount of labeled data from the target study areas for training (Nyborg et al., 2022). Unfortunately, generating labeled samples of cultivated crops is a time-consuming and expensive task in many areas across the globe (Wang et al., 2019; Belgiu et al., 2021). To address this challenge, transferring a model trained in an area with rich annotated labels, i.e. the source domain, to the unlabeled target study area, i.e. the target domain, has been investigated as a potential solution for crop mapping (Xu et al., 2020, 2021; Pelletier et al., 2019; Mohammadi et al., 2023; Nowakowski et al., 2021). Despite showing promising results, these methods often suffer from being localized (Wang et al., 2023, 2022), meaning that they are applied to target study areas with similar agroecological conditions to those of the source domain. Their generalization ability could be seriously challenged when applied directly beyond those study areas due to the discrepancies between the spectral and temporal properties of the source and target crops caused by different climatic and environmental conditions. Moreover, these discrepancies can also persist between the source and target crops across different years. It becomes difficult for the aforementioned methods to explicitly mitigate the domain shift problem in cross-regional and cross-time scenarios.

Unsupervised domain adaptation (UDA) methods have been employed in crop mapping and land cover mapping studies to reduce domain shift and improve knowledge transfer from a label-rich source domain to an unlabeled target domain (Nyborg et al., 2022; Wang et al., 2022, 2023). Nyborg et al. (2022) developed the TimeMatch UDA method capable of explicitly taking into account the temporal shifts that occur when crops are cultivated in different areas with varying climatic and agriculture management conditions. Wang et al. (2022, 2023) employed a feature extractor to project the source and target domain data to high-level features. Subsequently, their method minimized the maximum mean discrepancy metric to learn domain-invariant feature embeddings. Capliez et al. (2023) proposed a UDA method for land-cover mapping from image time series. Their method combined adversarial learning and self-training to progressively adapt a neural network from a source domain to a target domain.

Although previous studies have demonstrated the benefits of UDA methods, their limitation is that they require the source domain dataset during the adaptation phase. Unfortunately, there are scenarios where the labeled source samples cannot be employed and consequently, the discrepancy between the two domains cannot be resolved using the above-presented conventional UDA methods. This might happen due to computational resources and storage requirements (Yu et al., 2023). In the context of crop mapping, an increasing number of crop mapping datasets consisting of vast numbers of labeled samples are becoming progressively available. Since the ultimate goal of crop mapping studies is to develop large-scale mapping and monitoring applications (Boogaard et al., 2023; Rußwurm et al., 2023), it is required to leverage these extremely large labeled source datasets to achieve better results in a different target region or year. To this end, if the conventional UDA methods are employed, extremely large storage and high computational resources are required to store the massive datasets and train these methods on them (Fang et al., 2022; Kim et al., 2021). This can be excessively costly, particularly for organizations with budget constraints (Tuia et al., 2022). Moreover, training these deep learning-based methods on large amounts of data uses significant amounts of energy (Tuia et al., 2022). For example, researchers have approximated that the training process of GPT-3 (Brown et al., 2020) generates an equivalent of 552 tons of CO2, roughly equal to the emissions from six flights between New York and San Francisco (Patterson et al., 2021; Boudiaf et al., 2022). Therefore, it is high time to give due consideration to these financial costs and environmental consequences, topics that are often overlooked by the AI applications (Tuia et al., 2022).

Recently, there has been increasing interest in developing methods in which models are trained once on a large source dataset and subsequently can be adapted to as many target domains as desired using only the target datasets (Boudiaf et al., 2022). These methods, called source-free unsupervised domain adaptation (SFUDA), resolve the dependency of the conventional UDA methods on the source data and only require the target data during the adaptation stage (Kundu et al., 2020; Liang et al., 2020; Yang et al., 2022). In remote sensing, SFUDA has been successfully applied for image scene classification (Xu et al., 2022a) and hyperspectral image classification (Xu et al., 2022b). Moreover, source-free semi-supervised domain adaptation has also been used for land cover mapping, where some labeled samples are available in the target domain. For example, Lucas et al. (2021) used a CNN model trained on the source domain as a starting point and enhanced its performance in the target domain through additional training on the provided labeled target data. They used a novel regularizer during the target data training phase, which helps the model to gradually place more emphasis on knowledge derived from the target domain as the quantity of labeled target data grows.

Despite the advantages of SFUDA, they have not been developed for crop mapping. In this study, we proposed an SFUDA method for corn and soybean mapping across different regions and years in the U.S. using Landsat image time series in the presence of domain shift. We also assessed the transferability of our method across different countries and sensors using Sentinel-2 image time series collected from France. Moreover, we tested our method in more diverse agricultural areas in Denmark and France containing six classes. Many previous studies developed efficient methodological workflows to map crops at the end of the growing season. In contrast, a limited number of studies focused on identifying crops during the growing season (Cai et al., 2018; Khan et al., 2023; Lin et al., 2022; Rußwurm et al., 2023; Konduri et al., 2020), which is required for timely decision-making (Rußwurm et al., 2023). Therefore, the developed SFUDA method has been tested for both end-of-season and within-season crop type mapping. The main contributions of this study are as follows:

- We developed an SFUDA method for crop mapping based on mutual information loss, in which diversity and discriminability terms are balanced using a novel weighting strategy based on the mean confidence scores of the predicted categories.
- Our method eliminates the need for the source domain dataset during adaptation, leading to reduced financial and environmental costs compared to the conventional UDA methods. To the best of our knowledge, this is the first study focused on SFUDA for crop mapping.
- We demonstrated that our method is beneficial for both end-of-season and within-season crop mapping by conducting experiments across different regions and years and comparing our method to other UDA methods. Additionally, we demonstrated the effectiveness of our method across different countries and sensors.
- We verified that our method is model-agnostic and modular by employing CNN, LSTM, and Transformer, which are widely used in crop mapping.

2. The method

Our SFUDA framework (Fig. 1) consists of two stages, namely the source-free adaptation stage and the label prediction stage, given that a model pretrained on the source domain data is available prior to conducting SFUDA (Stage 0).

2.1. Stage 0: Pretraining stage

Stage 0 is performed prior to SFUDA and obtains an off-the-shelf source-pretrained model. We employed three main deep learning architectures used in crop mapping, namely CNN (Zhong et al., 2019; Pelletier et al., 2019), Transformer (Rußwurm and Körner, 2020; Garnot et al., 2022), and LSTM (Xu et al., 2020; Rußwurm et al., 2023)

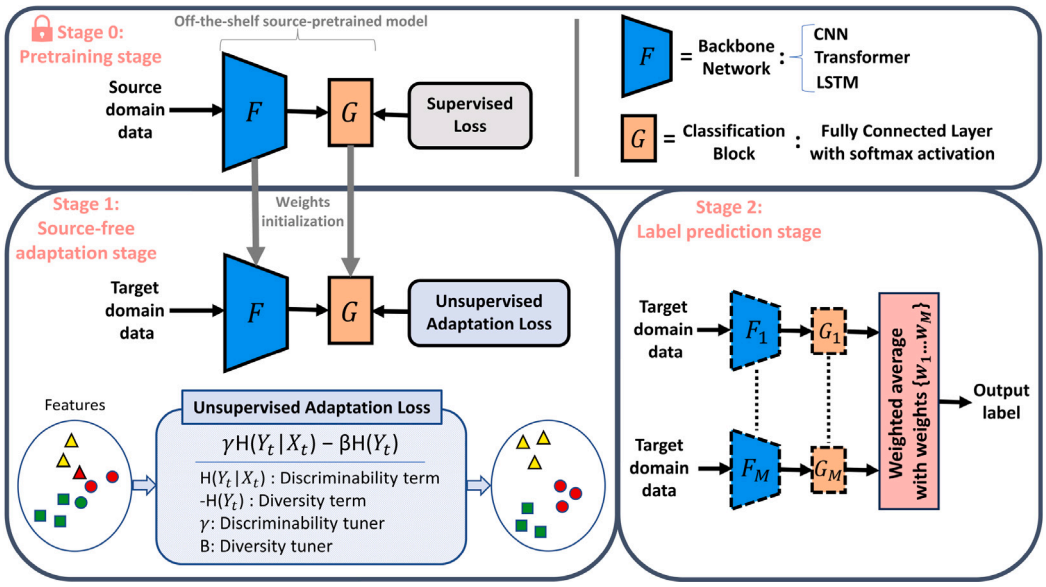


Fig. 1. Our framework for SFUDA for crop mapping, consisting of two stages: (1) source-free adaptation stage and (2) label prediction stage. Prior to performing SFUDA, it is assumed that we are given a source-pretrained model (Stage 0). Solid lines indicate that the network is being trained, while dashed lines indicate that the network is already trained during the previous stage and fixed in the current stage. Note that $\{F_1 ... F_M\}$ and $\{G_1 ... G_M\}$ denote the models obtained from training F and G with different γ and β values, and $\{w_1 ... w_M\}$ denote the weights obtained from the mean confidence scores generated by $\{G_1 ... G_M\}$. In the visualization of the features, different shapes correspond to different true classes while different colors represent different predicted classes.

as the backbone networks of the off-the-shelf source-pretrained model. Typically, these models are trained using a supervised loss function, such as the cross-entropy loss, on the source domain data (Xu et al., 2020, 2021; Pelletier et al., 2019; Turkoglu et al., 2021).

2.2. Stage 1: Source-free adaptation stage

If the off-the-shelf source-pretrained model is directly applied to the target data, it often shows poor performance due to the domain shift caused by temporal-spectral differences between the crops from the source and target domains (Wang et al., 2022). To mitigate the domain shift problem and improve knowledge transfer, Stage 1 employs an unsupervised loss function based on mutual information maximization, which achieved promising results in domain adaptation and few-shot learning in computer vision studies (Boudiaf et al., 2020; Veilleux et al., 2021; Liang et al., 2020).

Here, we define the unsupervised adaptation loss function used in Stage 1. Let X_t denote the random variable associated with the raw features for the target domain samples and Y_t denote the random variable associated with the target domain labels. The mutual information loss is defined as:

$$\mathcal{L}_{MI} := \mathcal{H}\left(Y_t \mid X_t\right) - \mathcal{H}\left(Y_t\right) \tag{1}$$

Minimizing this loss function is equivalent to maximizing the mutual information between $X_t \in \{x_i^t\}_{i=1}^{N_t}$ with N_t samples and their label predictions $Y_t \in \{c\}_{c=1}^{C}$ with C classes. L_{MI} consists of two terms: a conditional entropy term $\mathcal{H}\left(Y_t \mid X_t\right)$ and a marginal entropy term $\mathcal{H}\left(Y_t\right)$. Following Veilleux et al. (2021), these are obtained using α-entropy as:

$$\mathcal{H}\left(Y_t \mid X_t\right) = -\left(\frac{1}{\alpha - 1}\right)\left(\frac{1}{N_t}\right)\sum_{i=1}^{N_t}\sum_{c=1}^{C} p_{ic}^{\alpha} \tag{2}$$

$$\mathcal{H}\left(Y_t\right) = -\left(\frac{1}{\alpha - 1}\right)\sum_{c=1}^{C} \overline{p}_c^{\alpha} \tag{3}$$

where α is a hyperparameter and p_{ic} is the posterior distribution over labels given raw features, outputted by the G network. Let θ and ϕ denote the parameters of the F and G networks (Fig. 1), then $p_{ic} := \mathbb{P}\left(Y_t = c \mid X_t = x_i^t; \theta, \phi\right)$, and $\overline{p}_c$ is the marginal distribution over labels, i.e. $\overline{p}_c = \mathbb{P}\left(Y_t = c; \theta, \phi\right)$. Assuming $p_i := \mathbb{P}(X_t = x_i^t)$ denote the prior distribution over the target data samples, $\overline{p}_c$ is obtained as:

$$\overline{p}_c = \sum_{i=1}^{N_t} p_{ic} p_i \tag{4}$$

Since all target data samples are considered to have equal influence in the learning process, the prior distribution is uniform, i.e. $\{p_i\}_{i=1}^{N_t} = \frac{1}{N_t}$. Thus, $\overline{p}_c$ is computed as:

$$\overline{p}_c = \frac{1}{N_t}\sum_{i=1}^{N_t} p_{ic} \tag{5}$$

According to Eq. (1), minimizing $\mathcal{L}_{MI}$ results in decreasing the conditional entropy ($\mathcal{H}\left(Y_t \mid X_t\right)$) and increasing the marginal entropy ($\mathcal{H}\left(Y_t\right)$). If $\mathcal{H}\left(Y_t \mid X_t\right)$ is low, then X_t provides relevant information for predicting Y_t, meaning that after observing the values of X_t, there is a low uncertainty about the possible values of Y_t. On the other hand, $\mathcal{H}\left(Y_t\right)$ represents the inherent uncertainty or randomness in the variable Y_t without considering any other variable. If $\mathcal{H}\left(Y_t\right)$ is high, there is high uncertainty in Y_t, indicating a diverse range of predictions. Based on these explanations, reducing $\mathcal{H}\left(Y_t \mid X_t\right)$ encourages the model to generate confident predictions with high separability between different classes and unambiguous cluster assignments. This, however, may easily lead to non-suitable solutions, i.e. to mapping all samples into a single class. To mitigate this issue, $\mathcal{H}\left(Y_t\right)$ is maximized to encourage predictions to be diverse, thereby avoiding trivial solutions obtained when solely minimizing $\mathcal{H}\left(Y_t \mid X_t\right)$. Therefore, by minimizing $\mathcal{L}_{MI}$, $\mathcal{H}\left(Y_t \mid X_t\right)$ is minimized and $\mathcal{H}\left(Y_t\right)$ is maximized, which encourages discriminability and diversity of the predictions, respectively. Obtaining high performance in different target datasets requires different trade-offs between discriminability and diversity. Therefore, simple addition of $\mathcal{H}\left(Y_t \mid X_t\right)$ and $-\mathcal{H}\left(Y_t\right)$ can lead to performance degradation. To address this issue, we focused on balancing these two terms. Therefore, the unsupervised adaptation loss function in Stage 1 of our framework is defined as:

$$\min_{\theta,\phi}\{\gamma\mathcal{H}\left(Y_t \mid X_t\right) - \beta\mathcal{H}\left(Y_t\right)\} \tag{6}$$

where the hyperparameters γ and β are balancing weights assigned to the two entropy terms that can be viewed as the discriminability tuner and the diversity tuner, respectively.

An important question remains on how we can find optimal values for γ and β. Typically in non-UDA studies, the hyperparameters are tuned using the labeled source dataset as conducted in Xu et al. (2020, 2021) and Gallo et al. (2023). In UDA studies, this is done using either the labeled samples from the source domain (Wang et al., 2022, 2023; Long et al., 2018) or some labeled samples from a target domain (Nyborg et al., 2022; Fatras et al., 2021; Hoffman et al., 2018). We argue that all these procedures used to select the hyperparameters pose challenges. First, since there is no access to the source dataset in SFUDA, we cannot use the source dataset labels to tune the balancing weights assigned to these two entropy terms. More importantly, even if the source dataset were available in SFUDA, still using it for hyperparameter tuning would be suboptimal as the performance in the target domain is what we care about. This cannot be reflected by the success in the source domain due to the shift between the two domains (You et al., 2019). Second, using some labeled samples from the target domain for hyperparameter tuning is controversial to UDA and is in conflict with the application of UDA in real-world scenarios since it is assumed that the labels from the target dataset are not available. Besides, if some labeled target data are available, these should be used for semi-supervised domain adaptation rather than utilizing them for hyperparameter tuning since it can lead to improved results (You et al., 2019). These limitations call for more sophisticated procedures that can leverage the target domain data (not labels) in an unsupervised way for hyperparameter tuning. To address this gap, we proposed a novel hyperparameter tuning strategy, which is based on the idea that correctly classified samples tend to have more confident predictions than wrongly classified samples (Hendrycks and Gimpel, 2016).

Algorithm 1 shows the steps in the source-free adaptation stage. This stage consists of two main "for" loops (line 3 and line 16 of Algorithm 1). In the first loop (line 3 of Algorithm 1), we iterate over K distinct γ values while keeping β fixed at 1. In this way, we modify the weight attributed to the discriminability term in the unsupervised adaptation loss, while maintaining the diversity term weight at 1. In this loop, we train the backbone and the fully connected networks, initialized with the source-pretrained model weights, for one epoch (B batches) using the unsupervised adaptation loss with each γ value and $\beta = 1$ (from line 5 to line 7 of Algorithm 1). To determine which models to keep for the next stage, namely the label prediction stage, we check whether each predicted class encompasses at least 5% of the total sample count to ensure diverse predictions (line 9 of Algorithm 1). If this criterion is satisfied, we save the trained model alongside its mean confidence score of the predicted classes (line 11 of Algorithm 1). To compute the mean confidence score, we input all samples into the model and calculate the average of the maximum probabilities generated by the softmax output of the model (line 10 of Algorithm 1). In the second "for" loop (line 16 of Algorithm 1), we repeat the same procedure as the first loop, with the distinction that this time, we explore L different β values while keeping γ constant at 1. Consequently, this algorithm produces M trained models, each accompanied by its corresponding mean confidence score. These models and scores are saved (line 29 of Algorithm 1) for the subsequent label prediction stage.

Algorithm 1 SFUDA for crop mapping — Stage 1: Source-free adaptation stage, N_t denotes the total number of the unlabeled target samples, B denotes the number of batches in one epoch, K and L denote the number of the discriminability term ($\mathcal{H}\left(Y_t \mid X_t\right)$) and the diversity term ($-\mathcal{H}\left(Y_t\right)$) weights

Require: Source-pretrained model (F and G), unlabeled target samples ($X_t = \{x_i^t\}_{i=1}^{N_t} = \{x_b^t\}_{b=1}^{B}$), discriminability term weights ($\gamma = \{\gamma_k\}_{k=1}^{K}$), diversity term weights ($\beta = \{\beta_l\}_{l=1}^{L}$)

1: $m = 1$ ▷ Index for the model and its corresponding mean confidence score
2: $\beta = 1$
3: **for** $k = 1$ **to** K ▷ Loop for changing the discriminability term weights **do**
4: Initialize F_m and G_m using the source-pretrained model
5: **for** $b = 1$ **to** B ▷ Loop for training the model on all batches for one epoch **do**
6: Train F_m and G_m using x_b^t by optimizing $\gamma_k \mathcal{H}\left(Y_t \mid X_t\right) - \beta \mathcal{H}\left(Y_t\right)$
7: **end for**
8: $\{pr_i\}_{i=1}^{N_t} = \{\text{argmax}\left(G_m\left(F_m\left(x_i^t\right)\right)\right)\}_{i=1}^{N_t}$ ▷ Generating the predictions for all samples
9: **if** $(\frac{1}{N_t}\sum_{i=1}^{N_t} \mathbb{1}\{pr_i = 0\} > 0.05)$ **and** $(\frac{1}{N_t}\sum_{i=1}^{N_t} \mathbb{1}\{pr_i = 1\} > 0.05)$ **and** $(\frac{1}{N_t}\sum_{i=1}^{N_t} \mathbb{1}\{pr_i = 2\} > 0.05)$ ▷ Checking the diversity of the predictions (0,1, and 2 represent the class index in our datasets with three classes) **then**
10: $\bar{P}_m = \frac{1}{N_t}\sum_{i=1}^{N_t} \max\left(G_m\left(F_m\left(x_i^t\right)\right)\right)$ ▷ Calculating the mean confidence score
11: Save F_m, G_m, and $\bar{P}_m$ ▷ Saving the trained model and its corresponding mean confidence score if the diversity criterion is met
12: $m = m + 1$
13: **end if**
14: **end for**
15: $\gamma = 1$
16: **for** $l = 1$ **to** L ▷ Loop for changing the diversity term weights **do**
17: Initialize F_m and G_m using the source-pretrained model
18: **for** $b = 1$ **to** B **do**
19: Train F_m and G_m using x_b^t by optimizing $\gamma \mathcal{H}\left(Y_t \mid X_t\right) - \beta_l \mathcal{H}\left(Y_t\right)$
20: **end for**
21: $\{pr_i\}_{i=1}^{N_t} = \{\text{argmax}\left(G_m\left(F_m\left(x_i^t\right)\right)\right)\}_{i=1}^{N_t}$
22: **if** $(\frac{1}{N_t}\sum_{i=1}^{N_t} \mathbb{1}\{pr_i = 0\} > 0.05)$ **and** $(\frac{1}{N_t}\sum_{i=1}^{N_t} \mathbb{1}\{pr_i = 1\} > 0.05)$ **and** $(\frac{1}{N_t}\sum_{i=1}^{N_t} \mathbb{1}\{pr_i = 2\} > 0.05)$ **then**
23: $\bar{P}_m = \frac{1}{N_t}\sum_{i=1}^{N_t} \max\left(G_m\left(F_m\left(x_i^t\right)\right)\right)$
24: Save F_m,G_m, and $\bar{P}_m$
25: $m = m + 1$
26: **end if**
27: **end for**
28: M=m ▷ The number of saved models and their corresponding mean confidence scores
29: **return** $\{F_m\}_{m=1}^{M}$, $\{G_m\}_{m=1}^{M}$, $\{\bar{P}_m\}_{m=1}^{M}$

2.3. Stage 2: Label prediction stage

In this stage, the saved models along with their corresponding mean confidence scores obtained from the previous stage are utilized. To predict the target dataset labels, the model that resulted in the maximum mean confidence score could be used. However, this could lead to performance degradation as sometimes the model can generate wrong predictions with high confidence. Thus relying solely on the model that generated the maximum mean confidence score could be suboptimal. To mitigate this issue, we proposed a weighting strategy that leverages all mean confidence scores generated by different models trained with different γ and β pairs. Algorithm 2 shows the steps of this stage. We linearly map the mean confidence scores to values between 0.1 and 1 (line 2 of Algorithm 2) to determine the weights in our weighting strategy. Using these weights, we compute the weighted average of the softmax outputs of the models associated with these weights to obtain the final prediction for the target data (line 3 of Algorithm 2).

3. Study area and data preprocessing

3.1. U.S. datasets

To conduct crop mapping experiments, we selected three sites situated in IOWA (IA), Missouri (MO), and Mississippi (MS) states, referred to as Site IA, Site MO, and Site MS, respectively (Fig. 2). Each site covers 45 km × 45 km (1500 pixels × 1500 pixels), i.e. 202,500 ha. We chose study areas from both Dfa Cold and Cfa Temperate climatic zones (Beck et al., 2018) to map crops from different climatic zones

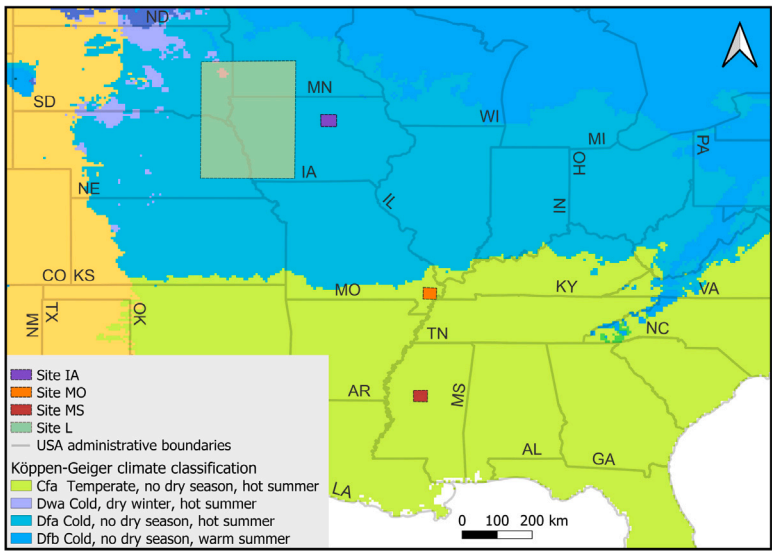


Fig. 2. The selected four study areas in the U.S. Their climatic zones (Beck et al., 2018) are also depicted in this map. These study areas are situated in six different states, namely Iowa (IA), Missouri (MO), Mississippi (MS), Minnesota (MN), South Dakota (SD), and Nebraska (NE).

Algorithm 2 SFUDA for crop mapping — Stage 2: Label prediction stage, N_t denotes the total number of the unlabeled target samples, M denotes the number of the saved models and mean confidence scores resulting from Algorithm 1

Require: Saved models ($\{F_m\}_{m=1}^{M}$ and $\{G_m\}_{m=1}^{M}$) and their corresponding mean confidence scores ($\{\bar{P}_m\}_{m=1}^{M}$) resulting from Algorithm 1, unlabeled target samples ($X_t = \{x_i^t\}_{i=1}^{N_t}$)

1: $w_{\min} = 0.1$, $w_{\max} = 1$

2: $\{w_m\}_{m=1}^{M} = \left\{ \frac{\bar{P}_m - \min\left(\{\bar{P}_i\}_{i=1}^{M}\right)}{\max\left(\{\bar{P}_i\}_{i=1}^{M}\right) - \min\left(\{\bar{P}_i\}_{i=1}^{M}\right)} \times \left(w_{\max} - w_{\min}\right) + w_{\min} \right\}_{m=1}^{M}$ ▷ Linearly mapping the mean confidence scores to values between 0.1 and 1 to obtain the weights for our weighting strategy

3: $\{y_i^t\}_{i=1}^{N_t} = \left\{ \frac{\sum_{m=1}^{M}\left(w_m \times G_m\left(F_m\left(x_i^t\right)\right)\right)}{\sum_{m=1}^{M} w_m} \right\}_{i=1}^{N_t}$ ▷ Generating the predictions by calculating the weighted average of the softmax outputs of the models using the weights obtained in the previous step

4: **return** $\{argmax(y_i^t)\}_{i=1}^{N_t}$

with the domain shift problem. We carried out the main analysis of our method using these three sites. Moreover, to demonstrate the effectiveness of our method on a larger spatial extent, we chose a site in the U.S., namely Site L (Fig. 2), which covers 450 km × 300 km (15000 pixels × 10000 pixels), i.e. 13,500,000 ha.

In the Analysis Ready Data (ARD) grid system, Site IA, Site MO, and Site MS are located in the center of the tile h18v07, the lower-right corner of the tile h20v11, and the center of the tile h20v14, respectively. Site L encompasses the entire six tiles of h16v06, h16v07, h16v08, h17v06, h17v07, and h17v08. More information about Landsat ARD tile grid system can be found at https://www.usgs.gov/media/images/conterminous-us-landsat-analysis-ready-data-ard-tiles.

To obtain the data for our experiments, we used Landsat C2 ARD surface reflectance composites (ARD, USGS, 2018) (publicly available on USGS's EarthExplorer web portal). We employed Landsat 8 Operational Land Imager (OLI) products and Landsat 7 Enhanced Thematic Mapper Plus (ETM+), containing non-spatially overlapping images at a spatial resolution of 30 m. Six optical bands, namely blue, green, red, near-infrared, shortwave infrared 1, and shortwave infrared 2 at each observation date were employed as the input of the crop mapping models.

In the selected study areas, corn and soybean are among the most frequent crop types. In our study, we defined corn, soybean, and the class Other (i.e., merged class of other land cover/land use types) as the target classes. The proportions of the target classes, calculated based on Cropland Data Layer (CDL), are given in Table 1.

CDL (publicly available on CropScape website portal) was used as the labels in our experiments, which is generated by the US Department of Agriculture (USDA) annually. CDL provides high mapping accuracy for corn and soybean classes (Xu et al., 2020; Cai et al., 2018; Wang et al., 2020; Xu et al., 2021). We provided the producer's and user's accuracy of CDL for these two classes in different sites and years in the Supplementary material. Based on the high accuracies of CDL for corn and soybean, many previous studies that aimed to map corn, soybean, and other classes used all the samples in CDL for testing the performance of the developed crop mapping methods (Xu et al., 2020; Cai et al., 2018; Wang et al., 2020; Xu et al., 2021; Zhong et al., 2014; Konduri et al., 2020). In line with them, we applied the same procedure. However, to further assess the effectiveness of our method, we conducted additional experiments using the samples selected based on the confidence layer associated with the CDL. This layer represents the predicted confidence of each pixel based on the ruleset used for classification. The additional experiments using these selected samples are provided in the Supplementary material.

Fig. 3 depicts the crop calendar of the selected study areas based on the state-level corn and soybean progress reports (Nass et al., 2010). Following Xu et al. (2020) and Xu et al. (2021), we defined the start of the overall growing season as the earliest date when planting of corn and soybean initiates across different states, and we defined the end of the overall growing season as the latest date when harvesting of these crops starts across different states. Accordingly, the overall growing season spans from March 17 to September 27. As the preprocessing method used in this study generates time series data with intervals of seven days, we used Landsat image time series from March 20 to September 25, covering a duration of 28 weeks. To preprocess the data, we followed the procedure proposed by Xu et al. (2020). Accordingly, a valid observation is defined as a pixel observation that is not labeled as filled, shadow, cloudy, dilated cloud, or unclear. Then, we selected the pixels that contain at least two valid observations in every third of the growing season as the valid pixels and filtered out the remaining pixels (invalid pixels) as they lack sufficient multi-temporal information. The statistics of valid pixels are shown in Table 1. Finally, we used linear interpolation to fill in the gaps caused by missing acquisitions and invalid observations, whose effectiveness was demonstrated in Xu et al. (2020) and Inglada et al. (2015). Therefore, our final preprocessed data contain pixels with 28 time steps at seven-day intervals from March 20 to September 25 and six bands.

Table 1
Statistics of valid pixels and different crop types in different sites and different years. IA, MO, MS, and L represent Iowa, Missouri, Mississippi, and the selected large study area, respectively. Site FR refers to a subset of samples in the PASTIS dataset (Garnot and Landrieu, 2021), collected from France. Since the dataset was already prepossessed by the authors, we did not apply further preprocessing, making the validity percentage 100%.

		Number of valid pixels	Validity percentage	Proportion of Corn	Proportion of Soybean	Proportion of Other
Site IA	2019	2,235,904	99.37%	54.50%	28.00%	17.50%
	2020	2,249,845	99.99%	55.73%	27.91%	16.36%
	2021	2,180,602	96.92%	50.60%	32.23%	17.17%
Site MO	2019	2,069,809	91.99%	13.65%	33.42%	52.93%
	2020	1,943,716	86.39%	13.42%	41.77%	44.81%
	2021	2,241,038	99.60%	16.50%	39.73%	43.77%
Site MS	2019	2,222,642	98.78%	12.68%	25.98%	61.34%
	2020	2,195,695	97.59%	10.35%	37.30%	52.35%
	2021	2,242,176	99.65%	13.35%	46.66%	39.99%
Site L	2021	145,729,426	97.15%	38.85%	32.55%	28.60%
Site FR	2018/2019	1,331,741	100%	54.96%	7.50%	37.54%

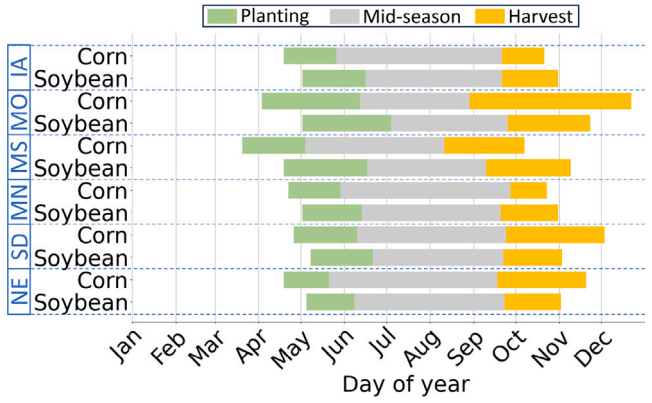


Fig. 3. Corn and soybean crop calendar in the selected study areas based on the state-level corn and soybean progress reports (Nass et al., 2010). These study areas are situated in six different states, namely Iowa (IA), Missouri (MO), Mississippi (MS), Minnesota (MN), South Dakota (SD), and Nebraska (NE).

In our cross-regional and cross-time tests using the U.S. datasets, the data collected in 2019 from different sites are the source domain datasets, and the data collected in 2020 and 2021 from different sites are the target domain datasets used to evaluate the crop mapping models. After the model was trained using the data of a site in 2019, it was tested on the same site in 2020 and 2021 in cross-time tests and tested on the other sites in 2020 and 2021 in cross-regional tests. The reason we used the data from two years as the target datasets is to demonstrate the consistency of the performance of our method in our experiments.

To illustrate the domain shift present in our data, we viewed the challenges faced in cross-regional and cross-time crop mapping from two perspectives (Wang et al., 2022). First, the phenology of the same crop can differ significantly across different regions and years due to variations in climatic conditions. We compared the temporal profiles of the classes in our datasets by investigating the mean NDVI temporal profiles in the upper section of Fig. 4. As an example, assuming that the model is trained on IA19 (Iowa 2019 data) and tested on IA21, MO21, or MS21, the temporal profiles of the same class (see Fig. 4) show the intra-class heterogeneity, hampering knowledge transfer from the source domain dataset to the target domain dataset.

Second, the temporal windows with large differences between the spectral values of different crop types play a significant role in crop mapping since they represent intervals when the classes are more distinguishable (Huete et al., 2002). However, the temporal discriminativeness patterns of the crop types vary across different regions and years (Zhao et al., 2020). The differences between the different pairs of mean NDVI temporal profiles are shown in the lower section of Fig. 4. For example, if the model is trained on Site IA19, it has learned to focus around week 16 for differentiating corn and soybean as this period contains a large difference between mean NDVI temporal profiles of corn and soybean. However, when this trained model is used to classify the crops in Site MS21, in the vicinity of week 16, corn and soybean do not have a large difference in terms of mean NDVI values. These crop types are more distinguishable in different periods (around week 11 and around week 24). These variations in temporal discriminativeness patterns and key phenological periods across different regions and years challenge transferring knowledge from the source domain dataset to the target domain dataset.

Another challenge associated with domain shift is the challenge of label shift, which occurs when the distribution of crop types in the target domain dataset differs from that of the source domain dataset. This has been observed to lower classification accuracy (Waldner et al., 2017). Table 1 shows that the proportions of the classes vary between different sites. For example, in site IA, Corn is the dominant class, while in site MO, the class Other is the dominant class. Therefore, our crop mapping experiments also entail the label shift challenge. Note that most of the sites contain over 96% valid pixels, as shown in Table 1. This indicates that the majority of pixels from these regions were included for training and accuracy assessment. Therefore, the imbalanced class distributions in our datasets largely reflect the real crop proportions in these areas.

3.2. France dataset

To demonstrate the effectiveness of our method across different countries and sensors, we used an additional study area for our experiments (see Fig. 5). We employed the PASTIS dataset (Garnot and Landrieu, 2021), which was collected using the Sentinel-2 satellite. Covering more than 4000 square kilometers, the PASTIS dataset was collected from four distinct regions in France, each characterized by diverse climates and crop distributions. It comprises 2,433 patches with a spatial size of 128 pixels by 128 pixels, being organized into five folds. The data was acquired between September 2018 and November 2019, with each patch encompassing 38 to 61 time steps. In total, there are 124,422 parcels, containing 18 different crop types and a background class representing non-agricultural land uses. We used corn and soybean samples from the first fold of the PASTIS dataset and randomly selected 500K samples from the remaining classes of the dataset. These randomly selected samples were then merged to form the class Other. The class distribution of this newly formed dataset, derived from the original PASTIS dataset, is presented in Table 1. To conduct the experiments using this dataset, we employed the same six optical bands used for the U.S. datasets, namely blue, green, red, near-infrared, shortwave infrared 1, and shortwave infrared 2.

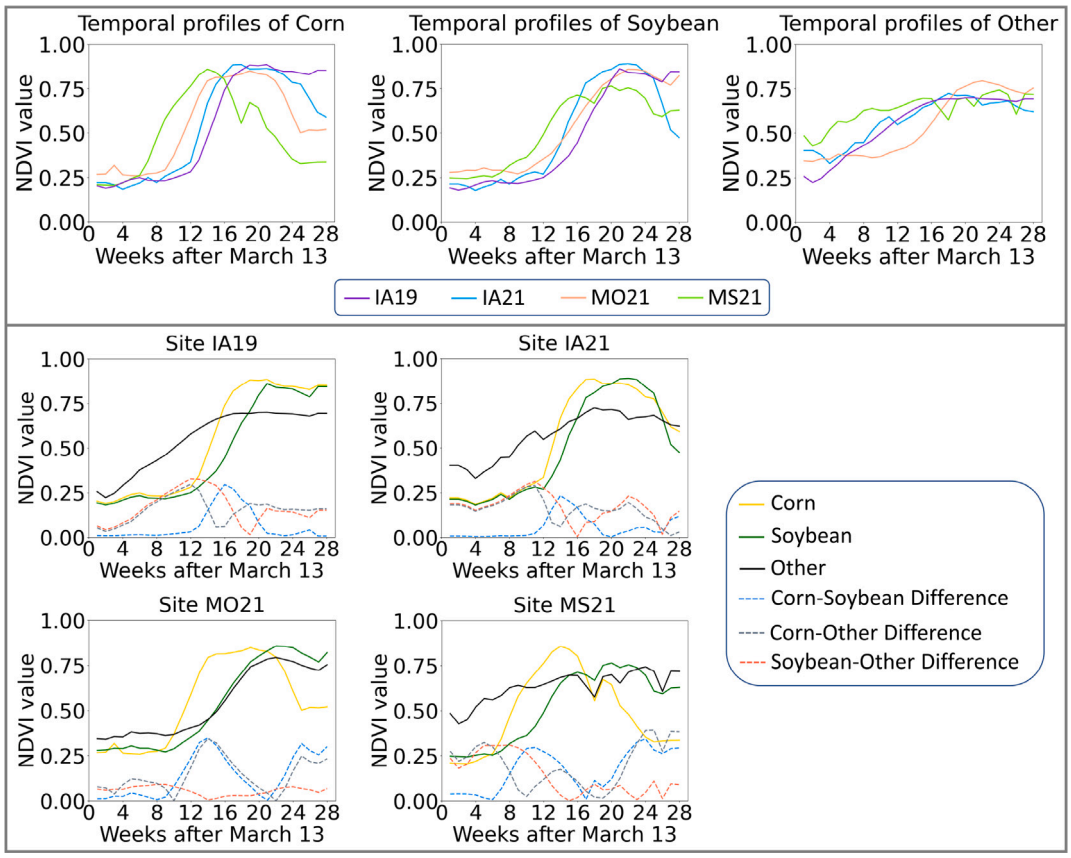


Fig. 4. Illustration of the domain shift problem in our study areas in the U.S. by using mean NDVI temporal profiles. The upper section shows the challenge of high dissimilarity between the temporal profiles of the same crop type grown in different years/regions. The lower section shows the challenge of high variance in the temporal discriminativeness patterns in different years/regions. Both challenges limit transferring knowledge from the source domain to the target domain.

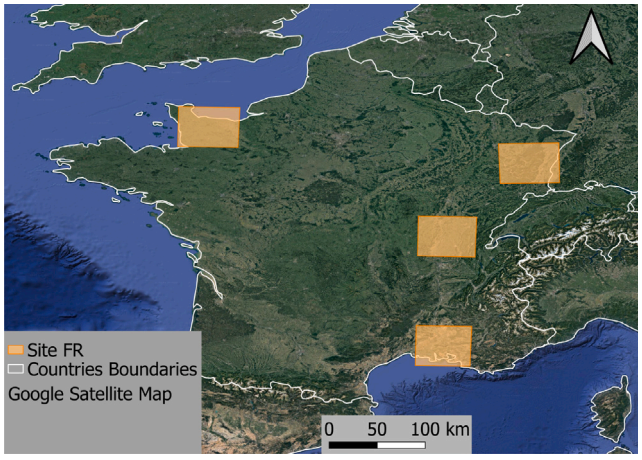


Fig. 5. Location of the study areas in the PASTIS dataset (Garnot and Landrieu, 2021) collected from France.

3.3. The TimeMatch dataset

To test our method on a more diverse agricultural dataset, we employed the TimeMatch dataset (Nyborg et al., 2022), designed specifically for domain adaption from satellite image time series. We used Sentinel-2 images available in the study areas from Denmark and France. Fig. 6 shows a map of the tiles 32VNH (Denmark) and 31TCJ (France), referred to as Site DK_TM and Site FR_TM, respectively. In the TimeMatch dataset, all available observations with cloud coverage $\leq 80\%$ and coverage $\geq 50\%$ between January 2017 and December 2017 were used. We selected five major crop types in these study areas, namely corn, meadow, winter barley, winter rapeseed, and winter wheat. Additionally, we randomly selected 100K samples from the remaining classes of the dataset and merged them to form the class Other. The class distributions of the dataset are shown in Fig. 6. All bands except the atmospheric bands B01, B09, and B10 were used in the dataset. The readers are referred to Nyborg et al. (2022) for more details about this dataset.

4. Experiments

4.1. UDA methods used for comparison to assess the efficiency of the proposed method

In end-of-season experiments, we compared our method with five conventional UDA methods and two SFUDA methods. These methods are listed below:

- **MMD** (Tzeng et al., 2014) minimizes the maximum mean discrepancy metric to learn domain-invariant feature embeddings. This method was adopted by Wang et al. (2022, 2023) for crop mapping from satellite image time series.
- **DAAN** (Ganin and Lempitsky, 2015) leverages a domain classifier and adversarial training with the aim of extracting domain invariant feature embeddings. This method was employed by Wilson et al. (2020) and Purushotham et al. (2016) to align the features extracted by temporal network architectures.
- **SpADANN** (Capliez et al., 2023) combines DANN and pseudo-labeling for land cover mapping from image time series.
- **CDAN+E** (Long et al., 2018) enhances DANN by conditioning the domain discriminator on classifier predictions and minimizing an entropy loss on target data.

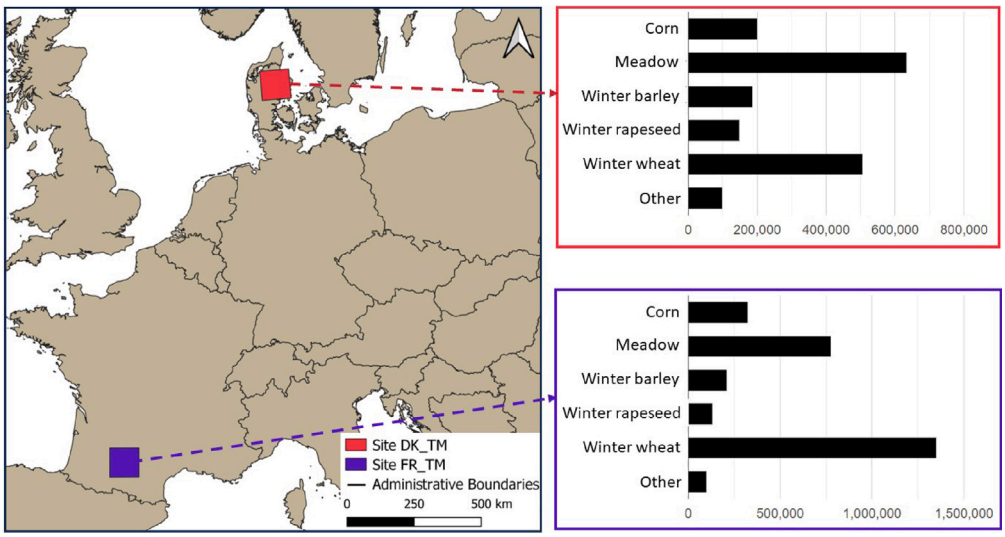


Fig. 6. The study areas and class distributions in France (Site FR_TM) and Denmark (Site DK_TM) derived from the TimeMatch dataset (Nyborg et al., 2022).

- **JUMBOT** (Fatras et al., 2021) minimizes the domain discrepancy of joint deep feature and label distributions by using mini-batch unbalanced optimal transport.
- **SHOT** (Liang et al., 2020), as an SFUDA method, first uses a mutual information loss to achieve domain alignment. Subsequently, it uses a pseudo-labeling strategy based on class centroids to augment the target representation learning and further improve the performance. It should be noted that our mutual information loss function is different from SHOT as we focus on balancing the discriminability and diversity terms using a novel weighting strategy, while SHOT uses a simple addition of these two terms.
- **AaD** (Yang et al., 2022) is an SFUDA method that encourages local neighborhood features in feature space to have similar predictions while features farther away in feature space have dissimilar predictions.

In within-season experiments, we compared our method to all the methods included in the end-of-season performance comparison. Furthermore, we incorporated an additional method, ELECTS (Rußwurm et al., 2023), which was specifically designed for within-season crop mapping. While ELECTS did not focus on within-season crop mapping in the presence of domain shift, our study investigated its performance in transfer experiments with the domain shift problem.

4.2. Implementation details

We implemented our framework with CNN, Transformer, and LSTM backbone networks in the PyTorch library using ITC Geospatial Computing Platform, equipped with an NVIDIA RTX A4000 GPU. In addition to the deep learning-based methods, we also included two conventional classification models in our comparison, namely Random Forest (RF) and Multilayer Perceptron (MLP), which were implemented using Scikit-learn and PyTorch libraries, respectively.

The CNN backbone in our framework consists of five 1D convolutional layers with 64,128, 256, 512, and 1024 filters with the filter size of 5, each followed by a Batch Normalization layer and a Relu activation. Then a Global Average Pooling layer was used as the output layer of this backbone network. The Transformer and LSTM networks in our framework were implemented based on Rußwurm et al. (2020) and Xu et al. (2020), respectively. More specifically, in the Transformer network, the dimension of the self-attention hidden feature, the number of self-attention heads, the number of self-attention layers, and the dimension of fully connected layer were set to 128, 2, 5, and 64, respectively. In the LSTM network, which is an attention-based bidirectional LSTM proposed by Xu et al. (2020), the number of hidden features and the number of layers were set to 256 and 2, respectively. Moreover, to implement RF and MLP, we also used the settings suggested by Xu et al. (2020). In RF, the number of trees and the number of features were set to 200 and 4, respectively. In the MLP network, four hidden layers were used with 256, 512, 512, and 256 neurons.

We performed Stage 0 in our study to obtain the source-pretrained model and mimic a source-free domain adaptation scenario using this off-the-shelf source-pretrained model. Typically, early-stopping using the validation data is a common approach to obtain the source-pretrained model (Caruana et al., 2000; Turkoglu et al., 2021; M Rustowicz et al., 2019; Mohammadi et al., 2021; Gadiraju and Vatsavai, 2023). To do so, we randomly selected 20% of samples of the source dataset for validation to perform early stopping and used the rest for training the models. In the pretraining stage, we used the Adam optimizer (Kingma and Ba, 2014) with a learning rate of 0.0001. In the source-free adaptation stage, we followed the training procedure used in Yang et al. (2022), setting the learning rate of the backbone network ten times smaller than the fully connected network, i.e., 0.0001 for the backbone network and 0.001 for the fully connected network. We set $\gamma = \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$ and $\beta = \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ as the discriminability tuner and the diversity tuner, respectively. α was set to 5. The code and data are available at https://github.com/ Sina-Mohammadi/SFUDA-CropMapping.

For the methods included in the comparison experiments, we enabled fair comparison by replacing their original feature extractor and classifier with our CNN backbone network and fully connected network. We initialized these networks using the source-pretrained model weights. The randomly selected 20% of samples of the source datasets were used for early stopping and hyperparameter tuning.

To evaluate the performance of the crop mapping methods in this study, we used macro $F1$ score formulated as:

$$\text{macro } F1 \text{ score} = \frac{1}{C}\sum_{i=1}^{C} F1_i \tag{7}$$

$$F1_i = 2\frac{P_i.R_i}{P_i + R_i} \tag{8}$$

where P_i and R_i are computed as:

$$P_i = \frac{TP_i}{TP_i + FP_i} \tag{9}$$

$$R_i = \frac{TP_i}{TP_i + FN_i} \tag{10}$$

In these formulas, $F1_i$, C, i, P_i, R_i, TP_i, FP_i, and FN_i denote the F1 score of the ith class, the number of classes, the index of the class, the precision of the ith class, the recall of the ith class, the true positives of the ith class, the false positives of the ith class, and the false negatives of the ith class, respectively.

4.3. End-of-season crop mapping results

4.3.1. Classification results in the U.S.

In this section, we compare the end-of-season results of the adapted models using CNN, Transformer, and LSTM backbone networks with those of the non-adapted models. The non-adapted models were trained on the source domain data and directly applied to the target domain data. Moreover, we compared these results with the results of RF and MLP.

The macro F1 scores in Table 2 show that the adapted models always outperformed the non-adapted models when using CNN as the backbone network. In the case of using the Transformer or LSTM as the backbone network, in most cases, our adaptation method improved the performance. In a few cases, however, the adaptation method achieved less accurate results. Nevertheless, the average performances in Table 2 show that the adapted models achieved better overall performance compared to the non-adapted models in both cross-regional and cross-time tests using the three backbones. Specifically, in cross-time tests, our adaptation method improved the performance by 12.9% (85.3% vs. 72.4%), 7.1% (80.0% vs. 72.9%), and 5.8% (88.3% vs. 82.5%) on average when using CNN, Transformer, and LSTM, respectively. In cross-regional tests, these improvements are 20.1% (66.2% vs. 46.1%), 12.5% (64.4% vs. 51.9%), and 8.8% (60.7% vs. 51.9%) for CNN, Transformer, and LSTM, respectively. To assess whether these improvements are statistically significant, we conducted two-sample t-tests comparing the macro F1 scores of the adapted models with those of the non-adapted models. The corresponding p-values are presented in Table 2. The table shows that in the cross-regional experiments, the performance improvements achieved by our adaptation method are statistically significant, with p-values below 0.05. In cross-time experiments, our adaptation method demonstrated a significant improvement in the results when utilizing the CNN backbone network. When employing the Transformer and LSTM backbone networks in cross-time experiments, although our adaptation method led to an increase in average macro F1 scores from 72.9% to 80% and from 82.5% to 88.3%, the associated p-values of 0.08 and 0.06 indicate that these improvements lack statistical significance. Furthermore, all adapted models outperformed RF and MLP in the vast majority of cases.

Additionally, the comparison between our adaptation method with a frozen and unfrozen CNN backbone network is presented in the Supplementary material. Results of adding vegetation indices to the Landsat bands in the cases where our adaptation method did not improve the performance are also reported in the Supplementary material. These vegetation indices include the Normalized Difference Vegetation Index (NDVI), Green Chlorophyll Vegetation Index (GCVI), and Enhanced Vegetation Index (EVI), which were also employed in a prior study (Wang et al., 2019).

To compare the classification maps generated by the non-adapted and the adapted models, we chose the results of the models with the CNN backbone network using the data collected in 2021. We show the classification maps and their comparison to the ground truth maps (error maps) using different possible pairs of source and target datasets in Fig. 7. Comparing the error maps of the adapted models with those of the non-adapted models shows the effectiveness of our adaptation method. This is consistent with the obtained scores in Table 2. Moreover, in Fig. 8, we show the classification maps, error maps, and F1 scores generated by the adapted and non-adapted models with the CNN backbone network using the data collected from the large study area. For this experiment, we chose the data collected from Site IA19 as the source data and tested the models on the data collected from Site L21 as the target data. The results demonstrate the effectiveness of our adaptation method, achieving a macro F1 score of 92.6%, which is superior to the non-adapted model results (89.2%). Considering each class, our adaptation method also showed superior performance with F1 scores of 93.6%, 93.6%, and 90.7% for corn, soybean, and the class Other. This improved performance is also reflected in the error maps, showing fewer misclassified pixels obtained by the adapted model.

Finally, the confusion matrices of the adapted and non-adapted models using the CNN backbone network along with the precision, recall, and F1 scores of individual classes for three transfer experiments are shown in Fig. 9. The confusion matrices of the adapted models show improved classification performance, obtaining higher scores in most cases.

4.3.2. Classification results across different countries and sensors

We conducted transfer experiments from the selected sites in the U.S. to the site in France (referred to as Site FR), and vice versa. The results reported in Table 3 show that our adaptation method improved the results of the non-adapted models in eight out of nine transfer experiments, demonstrating the effectiveness of our method across different countries and sensors.

4.3.3. Classification results in more diverse agricultural areas

We used the data from Denmark and France to perform transfer experiments in both directions: Denmark to France and France to Denmark. The results are presented in Table 4. When the Denmark data is the source dataset and the France data is the target dataset (DK_TM→FR_TM), our adaptation method improves the macro F1 score by 4.7% (39.5% vs. 34.8%). In the FR_TM→DK_TM transfer experiment, the macro F1 score increases from 25.5% to 33.9% using our adaptation method.

4.3.4. Comparison of the proposed SFUDA method with other adaptation methods for end-of-season crop mapping

We compared our method to five conventional and two source-free UDA methods in ten transfer experiments in Table 5, including FR→MO20, where our adaptation method was not able to improve the results of the non-adapted model. The table shows that our method outperformed the other methods in seven out of ten transfer experiments. It obtained an average macro F1 score of 65.9%, surpassing the second-best method, AaD, which obtained an average macro F1 score of 55.2%. The p-values from the two-sample t-test, as reported in the table, indicate that the performance difference between our method and the other methods is statistically significant. These improved results demonstrate the capability of our method to adapt to the target dataset using our novel adaptation method based on mutual information loss, in which diversity and discriminability terms are balanced using a novel weighting strategy. In two out of the three cross-sensor and cross-country transfer experiments, namely MS19→FR and FR→MS21, our adaptation method outperformed the other methods. In FR→MO20, all adaptation methods, including ours, could not improve the performance of the non-adapted models. However, our method still obtained more accurate results than the conventional UDA methods in this transfer experiment. These results highlight the advantage of our method over conventional UDA methods in a variety of domain shift settings, namely a different year, region, and sensor.

Table 2
Comparison of the end-of-season results of the adapted models with those of the non-adapted models as well as RF and MLP using the U.S. datasets in terms of macro F1 score. The numbers in red in each backbone category display the better results obtained by either the non-adapted model or the adapted model. The site and the year on the left side of the arrow (→) denote the source dataset, while those on the right side denote the target dataset. All values are reported in percentage. The p-values of two-sample t-test are reported to compare the macro F1 scores of the adopted models with those of the non-adapted models.

		CNN		Transformer		LSTM		RF	MLP
		Non-adapted	Adapted	Non-adapted	Adapted	Non-adapted	Adapted		
Cross-time	IA19→IA20	89.7	91.4	84.7	87.4	90.6	92.0	85.1	83.6
	IA19→IA21	90.2	94.1	91.3	92.6	92.9	94.3	83.9	86.4
	MO19→MO20	79.9	85.9	84.2	85.3	84.4	86.8	80.6	80.8
	MO19→MO21	59.2	81.8	57.3	77.3	68.3	87.0	65.8	63.5
	MS19→MS20	62.5	80.6	66.6	76.4	82.4	87.3	70.1	64.6
	MS19→MS21	52.8	77.8	53.3	60.7	76.5	82.6	60.1	60.5
	Avg.	72.4	85.3	72.9	80.0	82.5	88.3	74.3	73.2
	P-value	<0.05		0.08		0.06			
Cross-regional	MO19→IA20	30.8	60.3	31.3	61.5	39.0	52.2	36.7	34.8
	MO19→IA21	42.2	67.5	54.1	73.9	59.6	77.5	48.1	34.1
	MS19→IA20	29.4	48.6	21.6	42.9	19.8	19.8	34.7	26.6
	MS19→IA21	39.5	54.8	31.0	57.1	20.2	20.2	39.9	50.7
	IA19→MO20	58.0	63.4	52.9	50.9	61.4	62.7	48.7	51.5
	IA19→MO21	55.3	63.7	57.3	59.4	60.5	65.3	44.2	49.9
	MS19→MO20	63.5	80.9	71.4	75.7	67.3	73.4	57.3	65.4
	MS19→MO21	47.9	83.2	59.9	75.3	53.7	74.0	59.8	64.6
	IA19→MS20	50.9	76.3	65.6	66.5	57.6	69.5	42.0	44.1
	IA19→MS21	45.7	66.6	59.3	62.1	49.5	62.0	27.7	35.1
	MO19→MS20	44.7	55.2	62.1	74.3	69.4	79.4	55.9	54.5
	MO19→MS21	45.5	73.4	55.9	72.9	64.7	72.8	48.2	51.8
	Avg.	46.1	66.2	51.9	64.4	51.9	60.7	45.3	46.9
	P-value	<0.05		<0.05		<0.05			

Table 3
Comparison of the cross-country and cross-sensor end-of-season results of the adapted models with those of the non-adapted models using the CNN backbone network in terms of macro F1 score. The numbers in red display the better results obtained by either the non-adapted model or the adapted model. All values are reported in percentage.

	Non-adapted	Adapted		Non-adapted	Adapted		Non-adapted	Adapted
IA19→FR	40.6	47.0	FR→IA20	64.8	81.1	FR→IA21	54.4	80.9
MO19→FR	20.0	44.2	FR→MO20	56.2	54.5	FR→MO21	56.6	66.1
MS19→FR	21.2	39.6	FR→MS20	53.6	63.8	FR→MS21	41.7	61.9

Table 4
Comparison of the end-of-season results of the adapted models with those of the non-adapted models using the CNN backbone network and the TimeMatch dataset in terms of macro F1 score. The numbers in red display the better results obtained by either the non-adapted model or the adapted model. All values are reported in percentage. DK_TM and FR_TM refer to the subset of samples from the TimeMatch dataset that are collected from Denmark and France, respectively.

	Non-adapted	Adapted
DK_TM→FR_TM	34.8	39.5
FR_TM→DK_TM	25.5	33.9

4.4. Within-season crop mapping results

4.4.1. Within-season performances of the adapted and non-adapted models

In this section, we further demonstrate the advantages of our method by conducting within-season crop mapping experiments. We chose six U.S. transfer experiments and CNN as the backbone network and used the target datasets collected in 2021. The performance of the adapted models and the non-adapted models over time within the growing season starting from week 10 are shown in Fig. 10. As illustrated, the adapted models outperformed the non-adapted models in the vast majority of weeks, suggesting the benefit of our adaptation method for within-season crop mapping.

To determine the earliest classification date, we checked at which week the classification performance reaches 90% of end-of-season (week 28) performance, following the criteria for sufficient within-season performance suggested by Konduri et al. (2020). Fig. 10 demonstrates that the adapted models always reached sufficient within-season performance earlier than the non-adapted models, except for the MO19→IA21 experiment. Although in this experiment, both the adapted and non-adapted models reached sufficient within-season performance in the same week, the performance of the adapted model was higher in this week. In the other five transfer experiments, if our adaptation method is applied to the source-pretrained model, the earliest classification date reduces from 24 to 22, 23 to 20, 26 to 11, 27 to 10, and 26 to 10.

To visually inspect the changes over time in the classification maps generated by the non-adapted and adapted models within the growing season, we chose the transfer experiment IA19→IA21 with the CNN backbone network as an example and displayed the maps in Fig. 11. Comparing the error maps of the adapted models to those of the non-adapted models demonstrates the superiority of the adapted models, yielding fewer misclassified pixels.

4.4.2. Comparison of the proposed SFUDA method with other methods for within-season crop mapping

In this section, we compared our method to eight methods in within-season experiments. We trained the conventional UDA methods using two procedures. As these methods have access to the source datasets, they can be trained using either the full time series or the trimmed time series of the source datasets. In the latter approach, the time series were trimmed to match the number of available time steps in the target dataset time series.

Fig. 12 shows the comparison results over time within the growing season starting from week 10 in two transfer experiments. In MO19→IA21 experiment, our adaptation method and SHOT achieved comparable results from week 10 to 14, performing better than the competing methods, except for AaD performing slightly better in week

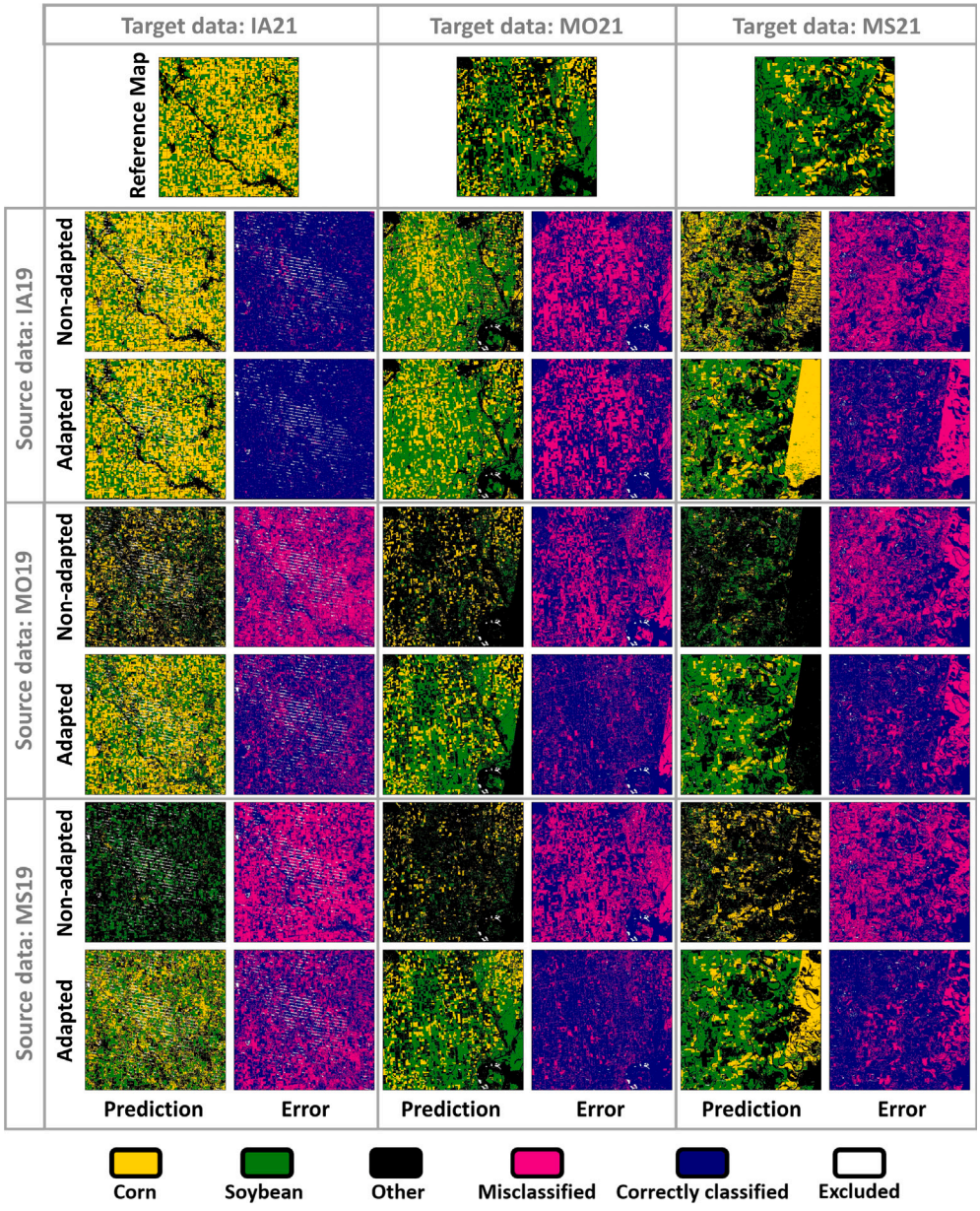


Fig. 7. The predicted maps and the error maps of the adapted and the non-adapted models using the CNN backbone network. The pixels that are in white are excluded from the dataset since they lack sufficient multi-temporal information. Note that each map covers 45 km × 45 km (1500 px × 1500 px) area.

Table 5
Comparison of our method to five conventional UDA and two source-free UDA methods using the CNN backbone network in terms of macro F1 score. The numbers in red display the best results. All values are reported in percentage. The p-values of two-sample t-test are reported to compare the macro F1 scores of our method with those of the other methods.

		Conventional UDA					Source-free UDA		
	Non-adapted	MMD	DANN	CDAN+E	JUMBOT	SpADANN	SHOT	AaD	Ours
IA19→IA21	90.2	93.1	92.4	93.8	89.1	91.5	93.8	88.3	94.1
IA19→MO21	55.3	48.3	53.0	58.3	55.5	56.1	63.0	56.4	63.7
IA19→MS21	45.7	46.2	46.5	61.9	49.4	26.3	59.1	57.4	66.6
MO19→MS21	45.5	75.8	58.5	62.4	73.2	32.8	68.1	48.3	73.4
MS19→IA21	39.5	46.3	39.6	40.0	37.0	32.9	38.3	61.1	54.8
MO19→IA21	42.2	53.8	50.9	61.0	58.5	59.9	50.3	53.8	67.5
MS19→MO21	47.9	75.6	78.4	71.2	77.0	42.7	62.5	61.4	83.2
MS19→FR	21.2	21.0	24.6	26.0	23.2	25.6	33.9	22.3	39.6
FR→MS21	41.7	29.6	27.8	22.4	43.0	31.7	11.1	47.7	61.9
FR→MO20	56.2	44.2	36.8	34.6	39.9	47.7	23.2	55.3	54.5
Avg.	48.5	53.4	50.9	53.2	54.6	44.7	50.3	55.2	65.9
P-value	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	

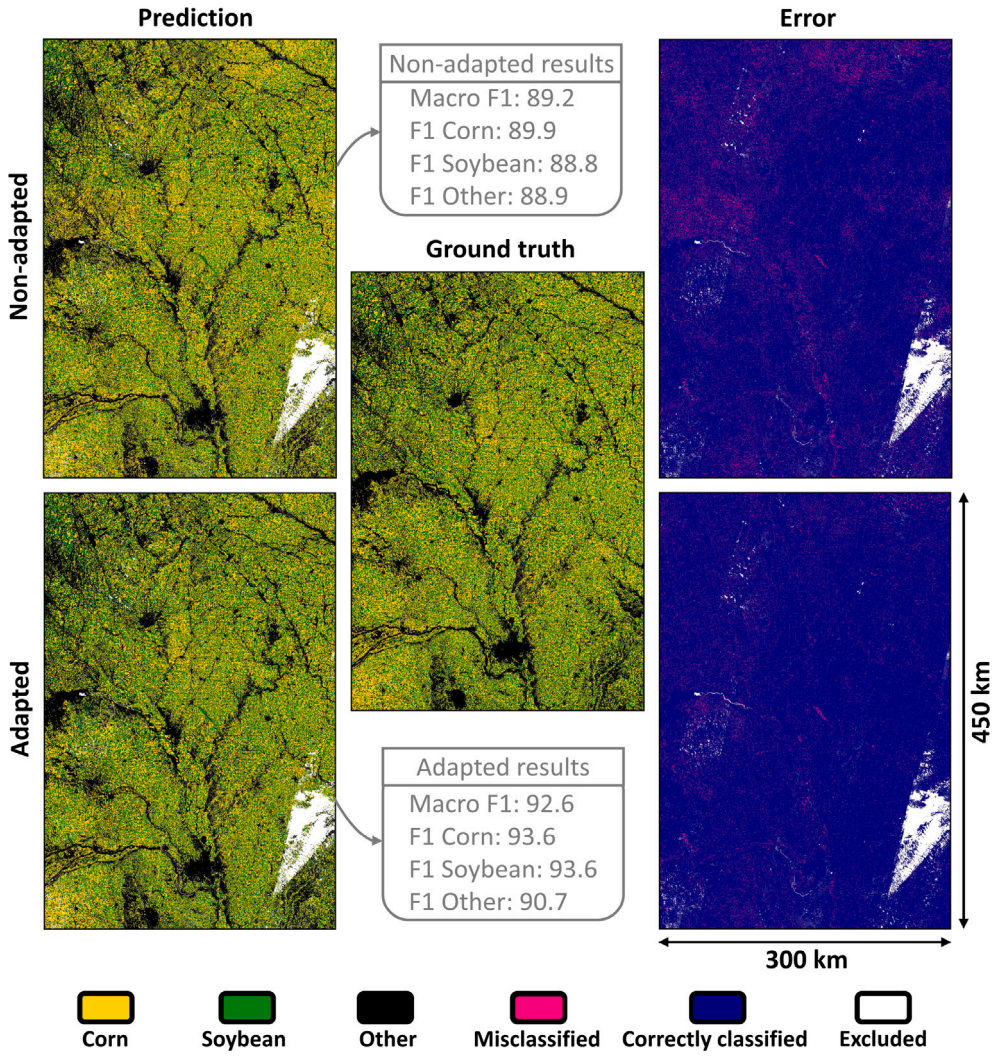


Fig. 8. The predicted maps, the error maps, and the F1 scores of the adapted and the non-adapted models with the CNN backbone network using the data collected from Site IA19 as the source data and the data collected from Site L21 as the target data. The pixels that are in white are excluded from the dataset since they lack sufficient multi-temporal information.

11. In the other weeks in this period, AaD obtained less accurate results, even compared to the non-adapted model in weeks 12 and 13. In weeks 15, 16 and 17, our method performed better than all competing methods. From week 18 to 22, our method obtained less accurate results compared to several methods. In weeks 23, 24, and 25, our method outperformed the other methods, while in weeks 26 and 27, AaD achieved the best results. In this transfer experiment, the performance of AaD fluctuated largely. It achieved highly accurate results in weeks 18, 21, 22, 26, and 27, while obtaining poor results in weeks 12, 13, and 15, performing worse than the non-adapted model. In this transfer experiment, it is not evident which method performs the best. To investigate this, a possible evaluation criterion could be taking the average from all the macro F1 scores obtained from week 10 to week 27. However, this procedure is not suited to within-season crop mapping as it does not take into account and reward the earliness of the predictions. For example, using this procedure, the macro F1 scores of week 12 and week 18 would have equal contributions to the final average score, while the former should have higher importance in within-season crop mapping. Therefore, we calculated the weighted average of the macro F1 scores using the weights determined by $1 - t/27$, which is defined based on the earliness reward (Rußwurm et al., 2023; Bilski and Jastrzębska, 2023). Here, t denotes the week number ranging from 0 to 27. In this way, the weights associated with the first week (index 0) and last week (index 27, end-of-season) equals 1 and 0, respectively. Table 6 reports these weighted average scores for different methods. As seen, our method obtained the highest score in MO19→IA21 experiment, surpassing AaD as the second-best method.

In MS19→IA21 experiment, from week 10 to week 18, our method, AaD, and SHOT achieved comparable performance, outperforming the other methods. From week 22 onwards, Aad achieved the best results. In terms of the overall performance, Table 6 shows that AaD achieved the highest weighted average score in MS19→IA21 experiment, performing better than our method, being the second-best method.

5. Discussion

In crop mapping, an increasing number of large labeled source datasets are becoming progressively available (Garnot and Landrieu, 2021; Turkoglu et al., 2021; Weikmann et al., 2021). This trend requires significantly increased storage capacity and the development of larger deep learning models to fit the potential complexity of the growing volume of labeled source datasets. Our study focused on developing an SFUDA method for the first time in the crop mapping literature. In contrast to the conventional UDA, this method does not require source domain data in the adaptation stage, and as a result, does not demand extremely large storage and high computational resources. This is beneficial, especially for research institutes and other organizations that have limited computational resources. Moreover, training deep learning

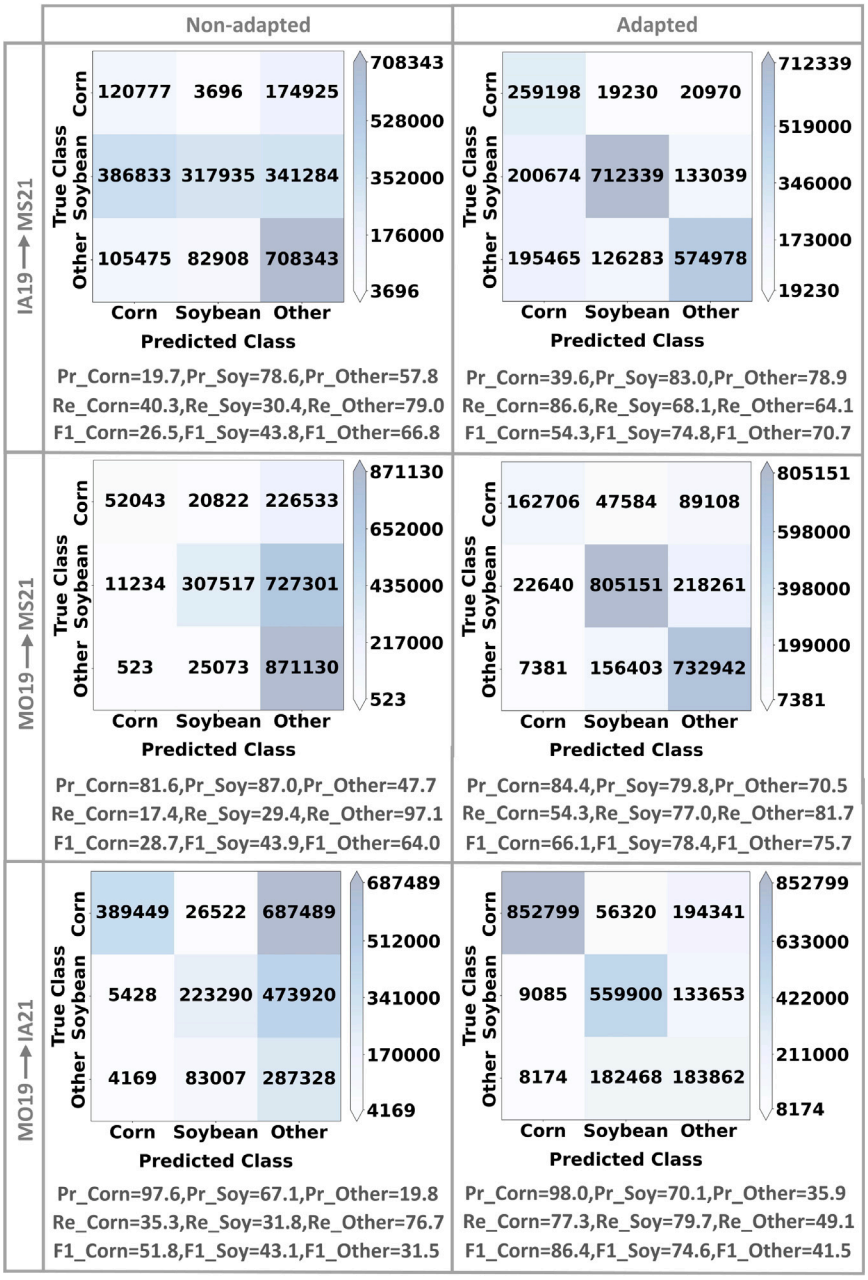


Fig. 9. The confusion matrices of the adapted and non-adapted models using the CNN backbone network along with the precision (denoted as Pr), recall (denoted as Re), and F1 scores of individual classes in three transfer experiments.

models on large amounts of data has environmental consequences, including increase of emissions and energy consumption (Tuia et al., 2022; Patterson et al., 2021; Boudiaf et al., 2022). The results achieved by our method in both end-of-season and within-season experiments demonstrated its advantage for transferring knowledge from the source dataset to the target dataset, being capable of tackling the domain shift problem.

5.1. Analysis of the obtained results

Table 2 showed that out of 54 transfer scenarios using the three backbone networks, our adaptation method improved the performance in 51 scenarios. This was achieved using the proposed weighting strategy. To demonstrate the advantage of this strategy in the label prediction stage (Stage 2 of our framework in Fig. 1), we compared its results with those obtained by the model with the maximum mean confidence score in Table 7. In the three chosen experiments, the two approaches achieved comparable performances in some cases. However, in some other cases, such as MO19→MO21 with the Transformer and LSTM backbone networks or MO19→MS21 with the LSTM backbone network, relying solely on the maximum mean confidence score resulted in poor performances, which are even worse than the performance of the non-adapted models (see Table 7). This happened because the model generated wrong predictions with high confidence in those cases. Therefore, our weighting strategy introduces a more reliable and effective method for the label prediction stage capable of tackling errors in the generated confidence scores.

To further demonstrate the advantage of the proposed weighting strategy, we compared its results with those obtained by the model trained with $\gamma = 1$ and $\beta = 1$ using the CNN backbone network in Table 8. In eight out of nine transfer experiments, the weighting strategy obtained superior results, highlighting the advantage of balancing the

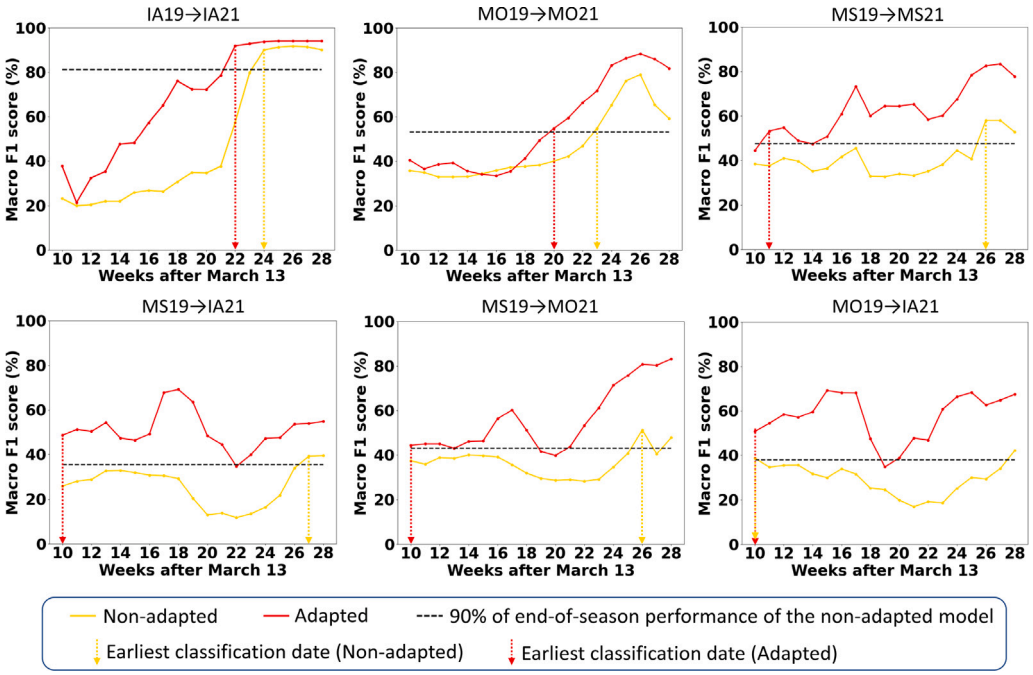


Fig. 10. Within-season performances of the adapted and non-adapted models using the CNN backbone network and the illustration of the earliest classification dates.

Table 6
Within-season performance comparison based on the weighted average of the macro F1 scores from week 10 to week 27 using the weights determined by considering the earliness reward. The numbers in red display the best results. All values are reported in percentage.

		MO19→IA21	MS19→IA21
	Non-adapted	30.5	26.5
Conventional UDA	CDAN+E	39.9	30.8
	CDAN+E (Trimmed)	40.9	34.1
	ELECTS	26.6	13.3
	SHOT	47.5	44.7
	DANN	39.5	28.4
	DANN (Trimmed)	40.0	30.9
	MMD	41.4	35.0
	MMD (Trimmed)	46.1	37.6
	JUMBOT	33.9	32.6
	JUMBOT (Trimmed)	43.0	39.5
	SpADANN	35.2	16.0
	SpADANN (Trimmed)	33.5	27.7
SFUDA	SHOT	47.5	44.7
	AaD	51.6	53.2
	Ours	56.2	51.7

diversity and discriminability terms over simply using them without weighting.

The confusion matrices in Fig. 9 demonstrate that the non-adapted models generated a considerable number of false positives for the class Other, mistakenly predicting many Corn and Soybean samples as the class Other. In comparison to the non-adapted models, our adaptation method notably reduced the number of false positives for the class Other, meaning that it successfully rectified a significant proportion of misclassifications where Corn and Soybean samples were inaccurately labeled as the class Other. These significant reductions in the number of false positives for the class Other and significant increases in the number of true positives for corn and soybean classes are important factors contributing to the large macro F1 score improvements, as reflected in Table 2.

Although our adaptation method resulted in the improvement of the most scores reported beneath the confusion matrices in Fig. 9, it degraded the recall score of the class Other. This is because compared to the non-adapted models, the adapted models generated a larger number of false negatives for the class Other, mistakenly predicting a larger number of the class Other samples as corn or soybean. Moreover, fewer true positives were generated for the class Other using our adaptation method. These factors led to recall score degradation for the class Other. But why did the adapted models generate a larger number of false negatives and fewer true positives for the class Other compared to the non-adapted models? To answer this question, we compare the confusion matrices of the adapted models to those of the non-adapted models and investigate how our adaptation method affects the classification of corn and soybean samples that were wrongly predicted as the class Other.

Investigating the confusion matrices of the non-adapted models shows that these models tend to predict a relatively larger number of samples as being the class Other as compared to the number of samples predicted as being corn and soybean. The temporal-spectral characteristics of corn and soybean in the source datasets are different from the target datasets due to the domain shift problem. Beyond the challenge of domain shift, the class Other, a combination of various categories, has high intra-class heterogeneity. Moreover, the composition of the class Other differs between the source and target datasets. These factors lead to less discriminative and more dispersed feature representations for the class Other. Consequently, the non-adapted models tend to predict numerous input samples as the class Other. This resulted in more true positives and fewer false negatives, increasing the recall score. On the other hand, this also resulted in a larger number of false positives for the class Other. This is why the non-adapted models obtained higher recall scores and lower precision scores for the class Other compared to the adapted models.

Investigating the confusion matrices of the adapted models shows that these models tend to largely affect the classification of the samples that were predicted as the class Other by the non-adapted models. Our adaptation method rectified the predictions of these samples by encouraging diversity and discriminability of the predictions. Despite rectifying many predictions of these samples by correctly classifying them as corn or soybean, it also wrongly assigned a subset of the class Other samples to either corn or soybean. This led to a reduction in the number of true positives and an increase in the number of false

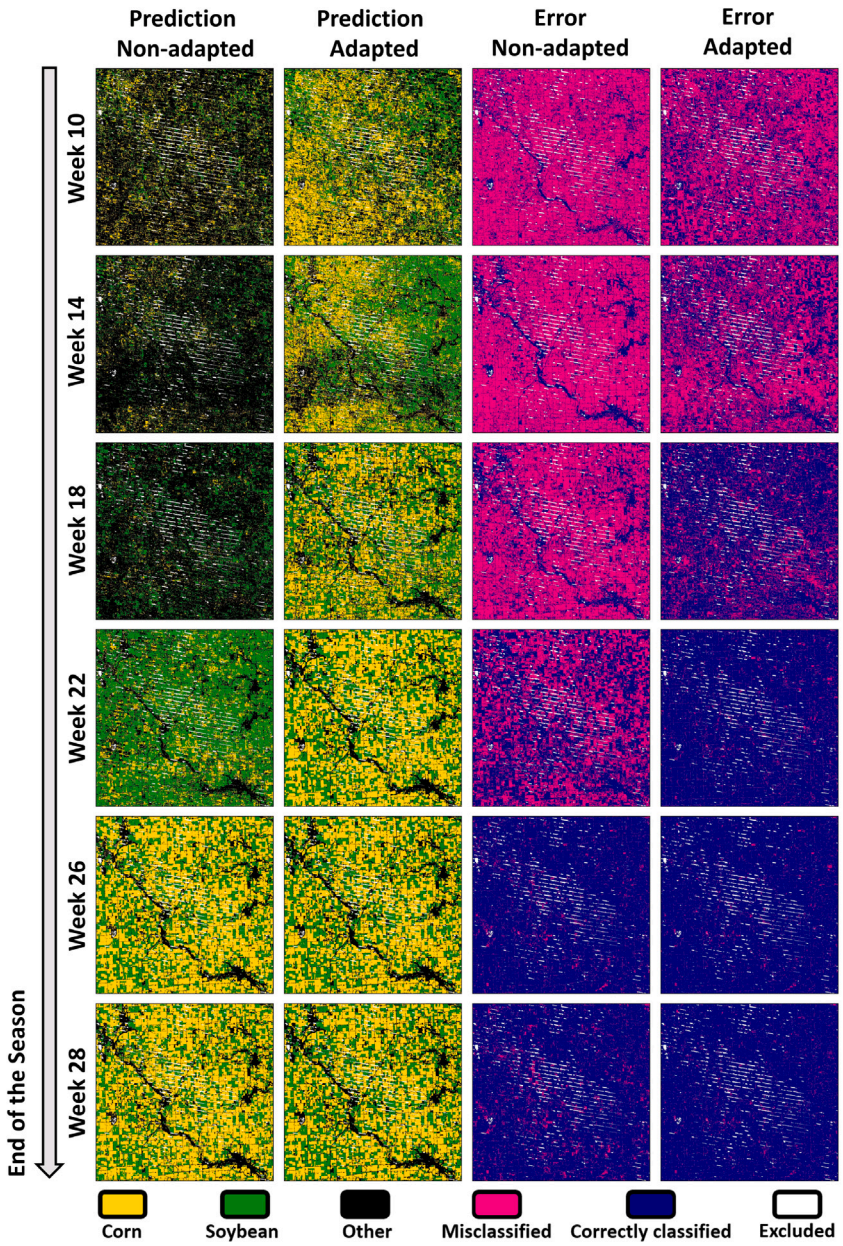


Fig. 11. The predicted maps and the error maps of the adapted models and the non-adapted model with the CNN backbone network over time within the growing season in the transfer experiment IA19→IA21. The pixels that are in white are excluded from the dataset since they lack sufficient multi-temporal information.

Table 7
Comparison of the results of our weighting strategy (denoted as Weighting) with the results obtained by the model with the maximum mean confidence score (denoted as Max).

	CNN			Transformer			LSTM		
	Non-adapted	Adapted (Weighting)	Adapted (Max)	Non-adapted	Adapted (Weighting)	Adapted (Max)	Non-adapted	Adapted (Weighting)	Adapted (Max)
MO19→IA21	42.2	67.5	69.4	54.1	73.9	57.0	59.6	77.5	77.8
MO19→MO21	59.2	81.8	81.0	57.3	77.3	57.0	68.3	87.0	58.9
MO19→MS21	45.5	73.4	70.7	55.9	72.9	73.3	64.7	72.8	55.4

negatives. Consequently, our adaptation method resulted in a lower recall score for the class Other compared to the non-adapted models.

To investigate how the adapted models affect the classification results in the feature space, we used t-SNE to visualize the features before the softmax layer at the output using 100K samples from the target data. We chose CNN as the backbone and three transfer experiments and show the results in Fig. 13. As illustrated, our adaptation method generated more diverse and discriminative predictions compared to the non-adapted models. This is also reflected in the Silhouette scores obtained by the adapted models. Ranging from −1 to 1, the Silhouette score measures how well the samples are clustered (Rousseeuw, 1987). Fig. 13 shows that the adapted models generated better-clustered features.

The results shown in Fig. 12 reveal that most adaptation methods performed better than the non-adapted models in within-season experiments in the majority of weeks. This emphasizes the effectiveness

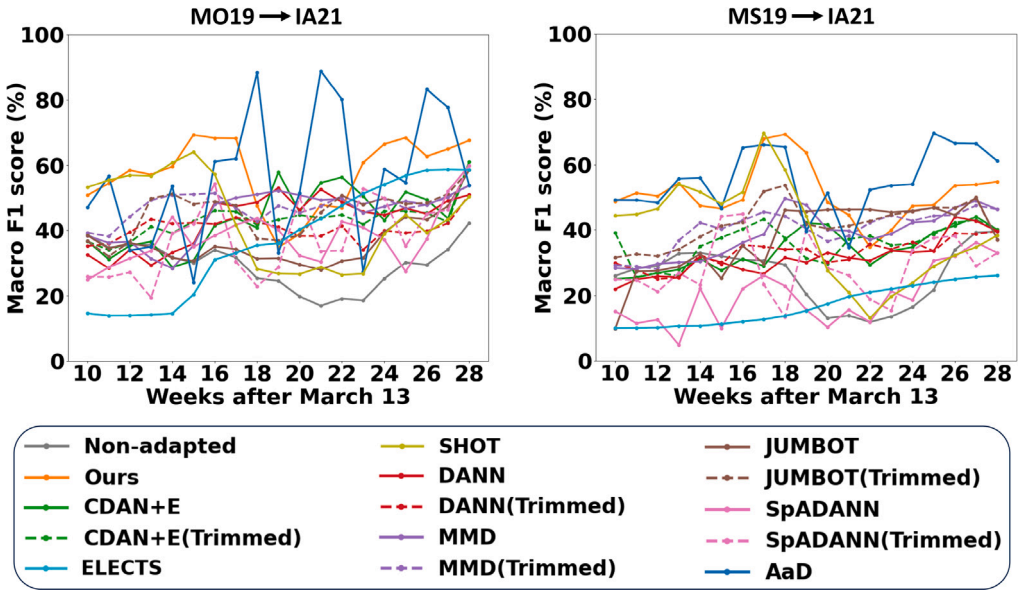


Fig. 12. Comparison of the proposed SFUDA method with other methods in two within-season transfer experiments. The conventional UDA methods are trained using either the full or trimmed time series. In the latter case, they are referred to as Trimmed, as shown in the legend of the figure.

Table 8
Comparison of the results of our weighting strategy (denoted as Weighting) with the results obtained by the model trained with $\gamma = 1$ and $\beta = 1$ using the CNN backbone network across nine transfer experiments. All values are reported in percentages, displaying macro F1 scores. The numbers in red display the best results.

	Non-adapted	Adapted (Weighting)	Adapted ($\gamma = 1$ & $\beta = 1$)
IA19→IA21	90.2	94.1	94.0
MO19→MO21	59.2	81.8	81.5
MS19→MS21	52.8	77.8	39.0
MO19→IA21	42.2	67.5	58.3
MS19→IA21	39.5	54.8	29.8
IA19→MO21	55.3	63.7	63.1
MS19→MO21	47.9	83.2	83.5
IA19→MS21	45.7	66.6	34.0
MO19→MS21	45.5	73.4	49.6

of UDA for within-season crop mapping, an aspect that has not been explored by other studies to the best of our knowledge. This finding advocates for the integration of UDA in methodologies tailored for within-season crop mapping, treating it as a domain adaptation challenge. This new perspective aligns with the idea that incomplete time series in within-season crop mapping can introduce greater disparities between the distributions of source and target datasets, justifying the application of UDA to enhance performance in such scenarios.

5.2. Limitations and future work

Despite the advantages of the presented SFUDA method for crop type mapping, our study has some limitations that can be addressed in future research.

First, as shown in Table 2, in one out of the total 54 transfer experiments using the U.S. datasets, the adapted model obtained less accurate results than the non-adapted model (in IA19→MO20 using the Transformer backbone network). This was also the case in one out of nine cross-country and cross-sensor transfer experiments shown in Table 3 (in FR→MO20). By means of inspecting the mean confidence scores and the corresponding test performances associated with the models trained with different γ and β values, we realized that the generated mean confidence scores did not represent the test performances well in these two cases. In other words, for most γ and β pairs, the models generated wrong predictions with high confidence scores that even our weighting strategy could not handle, unlike the other cases. We observed that if we had been able to identify the best-performing models in these two transfer experiments, we would have obtained macro F1 scores of 62.3%, and 68.3% instead of 50.9% and 54.5% in IA19→MO20 and FR→MO20, respectively, which are all superior to the scores obtained by the non-adapted models. Therefore, while our weighting strategy achieved promising results in the majority of transfer scenarios, future work can focus on proposing new unsupervised hyperparameter tuning methods to further minimize the errors in model selection. It is important to note that in the two transfer experiments in Table 2 where the adapted models obtained the same results as the non-adapted models, the prediction diversity criterion (lines 9 and 22 of Algorithm 1) was not satisfied, and therefore no adaptation occurred.

Second, we followed the experimental setting in other domain adaptation/transfer learning crop mapping studies that focused on identifying some classes (often major crop types) and merging all the remaining classes (often minor crop types) as the other/unknown class (Nyborg et al., 2022; Wang et al., 2022, 2023; Xu et al., 2020, 2021). Nevertheless, despite making up only a small portion of the total cultivated land, the minor crop types do bring ecological advantages to farmland, promote sustainability, and help ensure food and nutritional security, especially in the face of climate change (Waldner et al., 2019; Mayes et al., 2012). For example, to address the hidden hunger issue, it is required to have access to up-to-date and reliable information on agriculture production diversity, i.e., the number of crop types and their area cultivated in an agricultural area. This information is crucial for improving the dietary diversity of household diets and nutritional outcomes (Sibhatu et al., 2015). Calculating agricultural production diversity requires mapping not only major crop types but also minor crop types. This can bring some challenges, such as limited classification performance due to the lack of labeled samples and higher inter-class homogeneity and intra-class heterogeneity (Hamidi et al., 2021) compared to mapping only the major crop types.

Third, the purpose of pseudo-labeling and the conditional entropy term in the mutual information loss is similar as they both encourage the model to generate confident predictions with more discriminative features for each class and unambiguous cluster assignments (Berthelot et al., 2019; Boudiaf et al., 2020). Nevertheless, advanced pseudo-labeling techniques can be leveraged to help the model to generate

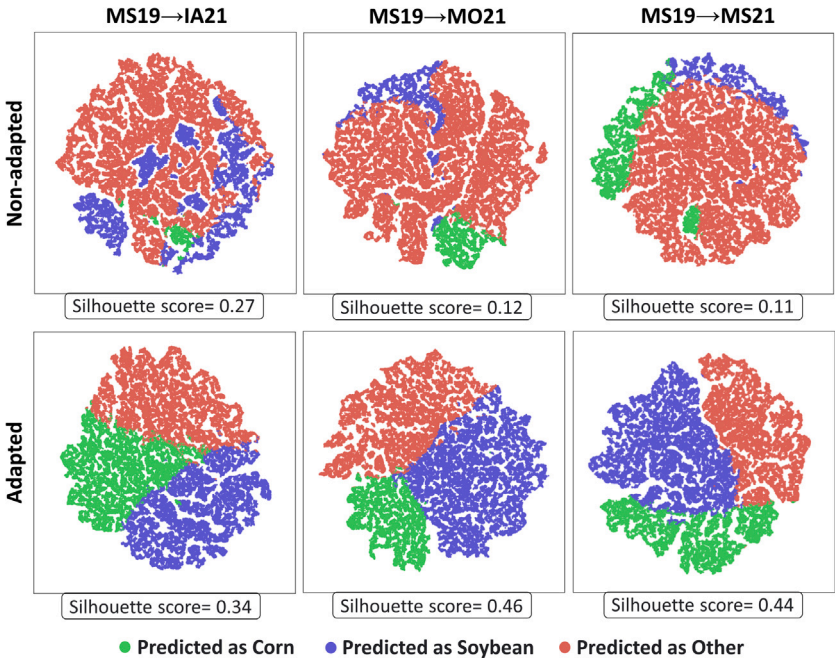


Fig. 13. Illustration of the effect of our adaptation method with the CNN backbone network on the output features by t-SNE. By employing t-SNE, a nonlinear dimensionality reduction technique, we reduced the original high-dimensional data to two dimensions for the purpose of visualization. The axes do not have a quantitative interpretation and do not correspond to any specific features or measurements from the original high-dimensional space due to the nonlinear dimensionality reduction.

more discriminative features. For example, after achieving domain alignment, Liang et al. (2020) used a pseudo-labeling strategy based on class centroids to augment the target representation learning and further improve the performance. Based on this, we encourage future work to add a subsequent pseudo-labeling step to our framework after achieving domain alignment to further improve the performance.

Fourth, the presence of clouds in optical satellite images makes crop mapping more challenging as it affects the integrity of the phenological information contained in image time series (Wang et al., 2022). To deal with this challenge, we followed the procedure introduced in Xu et al. (2020, 2021) to deal with the missing data. Future work can investigate other methods, such as fusing Landsat time series with MODIS or Sentinel-2 (Gao et al., 2017; Schreier et al., 2021) or reconstructing the images contaminated by clouds by employing image reconstruction methods such as generative adversarial network (GAN) (Zhou et al., 2022).

Fifth, the average cross-regional macro F1 scores are lower compared to the average cross-time macro F1 scores (Table 2) due to larger domain shifts in cross-regional transfer experiments, making crop mapping more challenging. Despite the effectiveness of our method for both cross-regional and cross-time crop mapping, more accurate methods are still required for practical applications. Therefore, our research can serve as a basis for future work dedicated to exploring and advancing domain adaptation methods to further alleviate the challenges in cross-regional experiments.

Finally, Landsat 7 ETM+ and Landsat 8 OLI instruments measure slightly different wavelengths, especially in NIR and SWIR 1, which can cause slight differences in the reflectance measurements (Roy et al., 2016; Flood, 2014). In our study, the experiments were conducted following the procedure in previous studies that combined Landsat 7 ETM+ and Landsat 8 OLI images without a wavelength-related preprocessing step (Wang et al., 2020). Future work could assess the impact of these differences on the classification results and accordingly mitigate the wavelength differences by employing an additional preprocessing step. For example, linear regression can be adopted to estimate Landsat 7 ETM+ reflectance from Landsat 8 OLI reflectance before being combined as employed in Flood (2014).

6. Conclusion

In this study, we developed an SFUDA method for cross-regional and cross-time crop mapping. The proposed method is more practical for crop mapping compared to the conventional UDA methods, which require high computational resources and data storage, leading to increased financial and environmental costs. The method is based on mutual information loss, in which diversity and discriminability terms are balanced using a novel unsupervised weighting strategy based on the mean confidence scores of the predicted categories. In the U.S. end-of-season experiments, where we focused on mapping corn, soybean, and the class Other, the adapted models with CNN, Transformer, and LSTM backbone networks performed better than their non-adapted counterparts. They improved macro F1 scores by 12.9%, 7.1%, and 5.8% on average in cross-time tests and by 20.1%, 12.5%, and 8.8% on average in cross-regional tests, respectively. Moreover, the adapted model with the CNN backbone network obtained a macro F1 score of 92.6% in a large study area covering 450 km × 300 km, surpassing its non-adapted counterpart with a macro F1 score of 89.2%. Our experiments, focused on mapping the same classes from Sentinel-2 image time series in France, demonstrated the efficiency of our method across different countries and sensors. We also tested our method in more diverse agricultural areas in France and Denmark with six classes. The results showed that the adapted models outperformed the non-adapted ones. Furthermore, in the within-season experiments, the adapted models achieved higher performance compared to the non-adapted models in the vast majority of weeks. The obtained results, along with their comparison to other UDA methods, showed the effectiveness of our method in both end-of-season and within-season crop mapping scenarios in the presence of domain shift between the source and target data. Moreover, the results demonstrated the modularity and adaptability of the method to different backbone networks.

CRediT authorship contribution statement

Sina Mohammadi: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Mariana Belgiu:** Writing – review & editing, Writing – original draft, Visualization, Supervision. **Alfred Stein:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and data availability

The code and data are available at https://github.com/Sina-Mohammadi/SFUDA-CropMapping.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.rse.2024.114385.

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